

Lecture 3: Option Pricing & A Two-State Tree Model

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Based on lecture notes by me and many previous CS 335 instructors

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Typical Financial Argument (“No Arbitrage”)

We'll argue that the portfolio's return must match the risk-free rate using a “proof” by contradiction...

Case 1: portfolio's return $>$ risk-free rate

This would create an *arbitrage opportunity*, according to the following strategy:

- a. Borrow cash from the bank at the risk-free rate.
- b. Construct the portfolio using that cash.
- c. After 3 months, sell the portfolio, repay the bank loan, pocket the difference (which is a profit, since the portfolio beat the risk free rate!).

An Arbitrage Opportunity

This is an arbitrage opportunity! What would happen if such an opportunity existed?

Many people would jump on it, and try to carry out that strategy. But then...

- a portfolio would be in demand
- b its price would quickly rise
- c portfolio's return would decrease, until it matches the risk-free rate again!

End of Lecture 2

Shorting the Portfolio

Case 2: Portfolio's Π 's return $<$ risk-free rate.

i.e., what if our risk-free portfolio were instead to get a lower rate of return than the risk-free rate?

Idea: Short the portfolio instead, to achieve another
arbitrage opportunity.

Definition: A short position is constructed by borrowing
assets we do not own (usually from a broker), selling
the assets for cash, and investing the cash.

However, we must deliver the assets back to the broker
in the future.

Mathematically, we own a negative quantity of the asset
(in this case, the portfolio).

Another Arbitrage Opportunity

This leads to a new strategy:

- Borrow the portfolio from a broker, then sell it for cash,
- a and invest the money in a bank account (which grows at the risk-free rate).
- b After 3 months, buy the portfolio back (from the stock market) using the cash from the bank, return the portfolio to the broker, and pocket the difference.

The difference is again a profit, since the bank cash earned interest at the risk-free rate, and (in this case) the portfolio's return was lower.

Another Arbitrage Opportunity

What happens if this arbitrage opportunity/strategy exists?

Again:

- lots of people want to carry it out.
- Many people would be trying to sell the portfolio (i.e., supply increase).
- Its price would quickly fall.
- The portfolio's return would increase to match the risk-free rate again.

No Arbitrage Principle: A portfolio which is risk-free must earn the risk-free rate of return.

(We will assume this throughout the course.)

Applying the No Arbitrage Principle

Let's now apply this concept to determine an appropriate value in our earlier (“two-state”) call option example, where $T = 3$, $K = \$21$:

Initially:	After $T = 3$ months:
$S(t=0) = 20$	$S(t=3) = \$22$, option value $V = \max(22 - 21, 0) = \$1$
	$S(t=3) = \$18$, option value $V = \max(18 - 21, 0) = \$0$

We will assemble a portfolio with a **guaranteed** value (at expire, $T = 3$ months).

Assembling A Risk-Free Portfolio

Construct the following portfolio Π consisting of two parts:

- A short position of 1 call option worth V .
- A long position in δ shares of the underlying asset (stock).

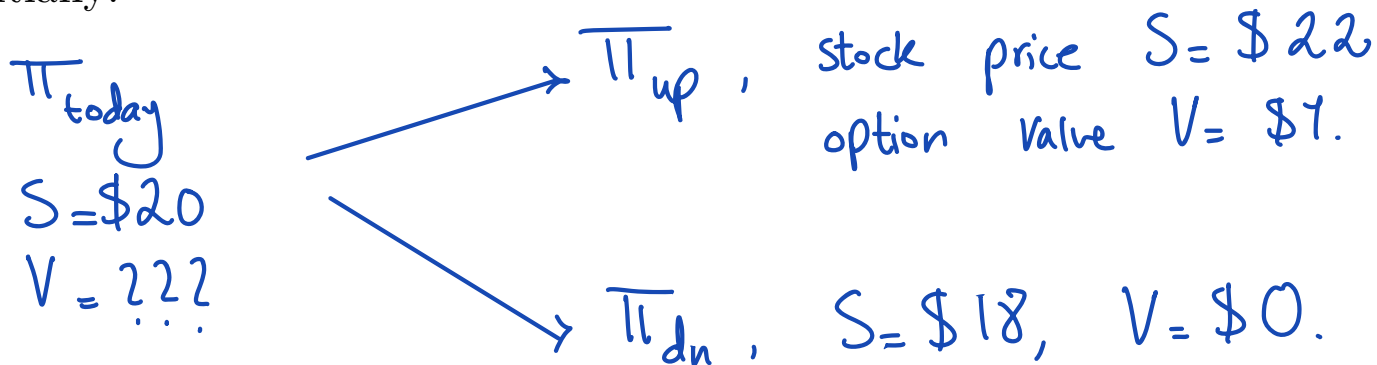
That is, the portfolio value is $\Pi = \delta S - V$.

Strategy: Choose the amount of stock shares δ such that the portfolio is risk free, i.e., has a guaranteed value in 3 months. Let's see how.

Assembling A Risk-Free Portfolio

What are the possible portfolio values in the two final states?

Initially:



Therefore:

"Up" case : portfolio value is $\pi_{\text{up}} = \delta S - V = (22)\delta - 1$

"Down" case : portfolio value is $\pi_{\text{dn}} = \delta S - V = 18\delta - 0$

To make it risk-free, just choose δ so that the "up" state and the "down" state have the exact same (final) portfolio value!

Eliminating Risk

Setting $\Pi_{up} = \Pi_{down}$ implies: $\delta(22) - 1 = \delta(18)$ which yields $\delta = 0.25$

Let's confirm the resulting portfolio values match:

- Up case $22(0.25) - 1 = 4.5$
- Down case $18(0.25) = 4.5$

Note: The portfolio is always worth 4.5 at $T=3$, regardless of what the stock does.

It's inherently risk free!

No Arbitrage Option Value

Our no-arbitrage principle says that such a risk-free portfolio *must* be earning the risk-free interest rate, so we *know* how fast its value is changing!

Let's say the risk-free rate is 12%. We can finally answer the question: What are the values of the portfolio and the option *today* (i.e., at $t = 0$)?

First, discount the portfolio value $\pi_{T=3} = 4.5$ back to today using the risk-free interest rate:

$$\pi_{\text{today}} = e^{-0.12 \times 0.25} (4.5) \approx 4.367$$

No Arbitrage Option Value

We know $\Pi_{today(PV)}$, $S(t=0)$, and δ , so we can determine the current option value V , using the definition of our portfolio value Π .

$$\Pi = \delta S - V, \quad \text{implies} \quad V = \delta S - \Pi = (0.25)(20) - 4.367 = 0.633$$

Therefore the value/price of the option today should be \$0.633.

This is called the no arbitrage value or fair value of the call option.

What about our deal?

We've shown that a smarter price/value for the option, today, is \$0.633. This is significantly more than the \$0.097 we found before when using expected discounted cash flow (rather than no arbitrage pricing).

Earlier, the sneaky banker offered you \$0.30 for the option (which perhaps seemed unwise at the time!)

However, our no arbitrage pricing analysis has confirmed that ~~I should be able to earn my claimed profit of~~

So what is the actual strategy to do so?

Let's work this out...

$0.633 - 0.3 = \$0.33$ per option.

Exploiting a Lower Option Value

What if $V < 0.633$?

Let $V^* < 0.633$ be the option price (e.g., 0.30 as I offered).

Strategy:

- Short 8 shares of stock and buy one call option.

$$\therefore \text{cash} = 8S - V^* = \left(\frac{1}{4}\right)(20) - (0.3) = 4.7$$

- Deposit cash in bank (to earn interest).

- After 3 months, cash in bank

$$= 4.7 \cdot \exp\left(0.12 \times \frac{3}{12}\right) = B$$

Exploiting a Lower Option Value

Two states to consider:

$S \rightarrow 22$: - Earn 1 dollar from option. So we have
cash = $B + 1$.

$$\text{- stock owed} = \left(\frac{1}{4}\right)(22) = 5.5$$

$$\text{- net gain} = B + 1 - 5.5 = B - 4.5$$
$$= 4.7 e^{0.12 \times 0.25} - 4.5$$

$$4.5 e^{-0.12 \times 0.25}$$

$$\approx 4.367$$

$$= e^{0.12 \times 3/12} (4.7 - 4.367)$$

$$= e^{0.12 \times 3/12} (0.333)$$

Exploiting a Lower Option Value

$$V - V^* = 0.633 - 0.30 = 0.333, \quad \text{plus 3 months interest.}$$

$S \longrightarrow 18$: - option value yields nothing (0).
So we have B.

$$\text{- stock owed} = \left(\frac{1}{4}\right)(18) = 4.5$$

$$\text{- net gain} = \boxed{B - 4.5}$$
$$= e^{0.12 \times (3/12)} (0.333)$$

Either way, we profit. The present value of our net gain is $0.333 = V - V^*$

Additional Notes

End of Lecture 3

In general, if we can buy 1 option for some $V^* < V_{noarbitrage}$ we can lock in a guaranteed profit of $V_{noarbitrage} - V^*$.

EXERCISE:

Work out the analysis for the case where the proposed option value is instead greater than 0.633. (How do you still guarantee a profit?)

Note: The no-arbitrage price is independent of the actual probabilities of the up and down movement of stock! Likewise, our choice of δ had no dependence on these probabilities.

The preceding is a hedging strategy, because we have used the option and the stock to minimize/avoid risk due to the stock movement.